

Continuous-Time Bandpass Filter Design, Modeling & Analysis Tool on MATLAB

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Outline

- **Introduction**
- **Motivation**
- **Tool Overview**
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- **Conclusion**
- **Key References**
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Introduction: Continuous-Time Filters

Fundamental Building Blocks:

- Essential for modern analog and mixed-signal systems (RF transceivers, high-speed data converters etc).
- Crucial in applications where sampling rates for discrete-time (switched-capacitor) or digital filters are prohibitively high.

Inherent Design Challenges:

- Highly sensitive to component variations and the non-idealities of active devices.
- Generally requires some sort of tuning to calibrate the taped-out chip.

The Design Bottleneck:

- Initial synthesis typically relies on ideal mathematical models.
- Severe performance deviations are often only discovered later during transistor-level simulations.
- This discrepancy forces a tedious, highly iterative re-design loop, especially compared to well-established discrete-time methodologies.

Motivation: Creating a Bandpass Filter Design Tool

Design Complexity:

- Bandpass filters are highly sensitive to component variations and parasitic poles.
- Calculating exact coefficients for accurate pole placement to achieve the desired Q-factor and center frequency is mathematically intensive.

Circuit Non-Idealities:

- Real OPAMPs exhibit finite open-loop DC gain, causing leakage when used as integrators.
- Leakage and other non-idealities cause significant bandwidth widening and a reduction in the effective Quality Factor.

The Implementation Chasm:

- Transitioning directly from abstract analysis to transistor-level implementation often leads to costly design reiterations.

Proposed Solution:

- A unified MATLAB environment bridging mathematical synthesis and implementation.
- Enables rapid “what-if” analysis of circuit non-idealities.
- Generates a Verilog-A model to serve as a reference benchmark testbench in Cadence.

Tool Workflow Overview

1. Coefficient Synthesis

Calculates transfer function coefficients based on target center frequency (f_0).

2. Non-ideality Modeling

Calculates Q-factor and BW while incorporating finite DC gain (A_{vol}) to extract a realistic transfer function.

3. Simulink Model Generation & Verification

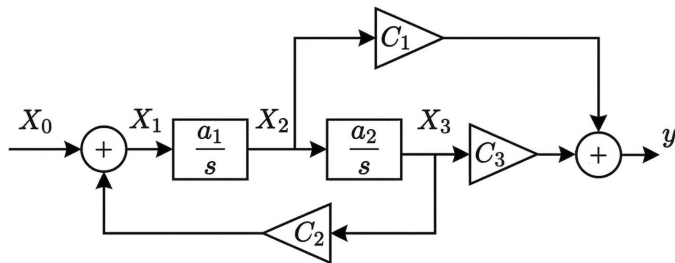
Auto-generates fully wired Simulink models to verify Transient and AC sweep behaviors.

4. Verilog-A Export

Exports current-mode integrator & bandpass filter models as Verilog-A behavioral code to allow integration into Cadence Virtuoso or any other design environment.

2nd-order Continuous-Time Bandpass Filter Architecture

- This block diagram illustrates the Continuous-Time Cascade of Resonators Feedforward (CRFF) 2nd-order resonator architecture.
- The configuration implements a two-integrator loop connected with a feedback path.
- The negative feedback loop shifts the open-loop poles from DC to the target center frequency on the imaginary axis.



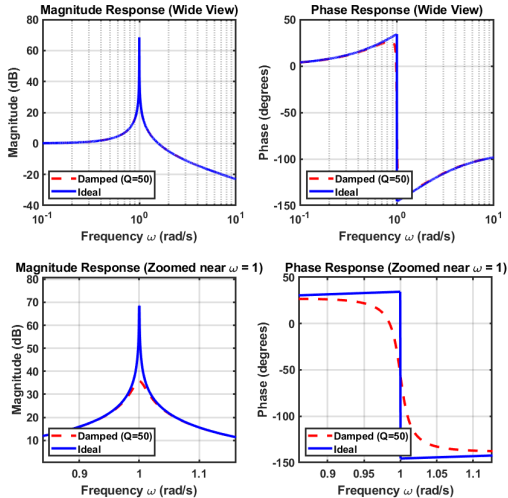
$$H(s) = \frac{a_1 C_1 s + a_1 a_2 C_3}{s^2 - a_1 a_2 C_2}$$

Assuming integrators with
unity gain ($a_1 = a_2 = 1$)
 $C_1 = 0.69$, $C_2 = -1$, $C_3 = 1.01$

$$H_{ideal}(s) = \frac{0.69s + 1.01}{s^2 + 1}$$

$$H_{damped}(s) = \frac{0.69s + 1.01}{s^2 + 0.02s + 1}$$

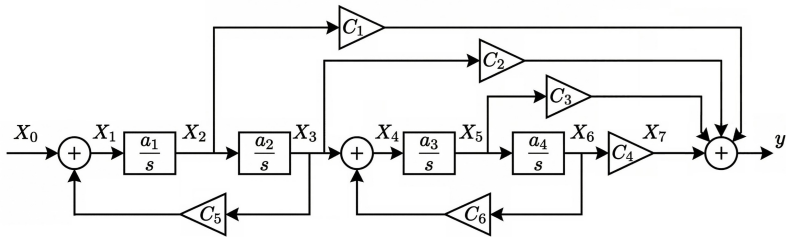
Damping Analysis: 2nd-order Filter Response



- AC response characterizes the stability constraints of the 2nd-order resonator.
- **Ideal Target ($Q = \infty$):** Achieves a sharp filtering notch but operates at the edge of marginal stability.
- **Damped Design ($Q = 50$):** Introduces a damping factor to establish unconditional system stability.
- This controlled damping trades a reduction in peak gain to ensure stable operation.
- Poles of the ideal system:
 - $0.00000000000000 \pm 1.00000000000000i$
- Poles of the damped system:
 - $-0.01000000000000 \pm 0.999949998749937i$

4th-order Continuous-Time Bandpass Filter Architecture

- High-resolution applications may require a higher order filter configurations.
- This block diagram details the architectural structure of the CRFF 4th-order filter.
- Feedforward paths route current signals directly to the final summing node.



$$H(s) = \frac{C_1 s^3 + C_2 s^2 + (C_3 - C_1 C_6)s + (C_4 - C_2 C_6)}{(s^2 - C_5)(s^2 - C_6)}$$

$$C_1 = C_4 = 0.3$$

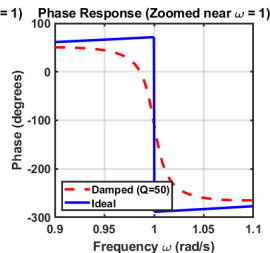
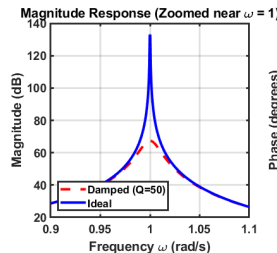
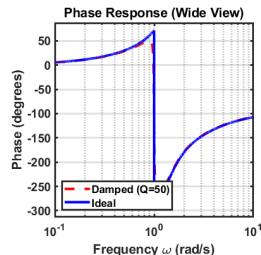
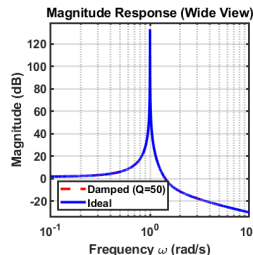
$$C_3 = C_2 = 0.9$$

$$C_5 = C_6 = -1$$

$$H_{ideal}(s) = \frac{0.3s^3 + 0.9s^2 + 1.2s + 1.2}{s^4 + 2s^2 + 1}, \quad H_{damped}(s) = \frac{0.3s^3 + 0.9s^2 + 1.2s + 1.2}{(s^2 + 0.02s + 1)^2}.$$

Damping Analysis: 4th-order Filter Response

- AC magnitude and phase response curves characterize the stability boundaries of the 4th-order filter network.
- Poles of the ideal system: ($Q = \infty$)
 - $-0.000000000005959 \pm 1.0000000008852252i$
 - $0.000000000005959 \pm 0.999999991147748i$
- Poles of the damped system: ($Q=50$)
 - $-0.010000009320415 \pm 0.999949998700570i$
 - $-0.009999990679585 \pm 0.999949998799306i$



Custom Normalized Transfer Functions

- The tool allows user to enter custom transfer functions while it adjusts internal stage quality factors to ensure the global system BW adheres to user specs.
- By default, this setting is disabled and the tool uses precalculated normalized transfer function coefficients for all orders.

Default Coefficients for 2nd & 4th-order ($a_n = 1$)

The 2th-order default tool setting synthesizes with the feed-forward coefficients below:

$$c_1 = 0.69, \quad c_2 = -1, \quad c_3 = 1.01.$$

The 4th-order default tool setting synthesizes with the feed-forward coefficients below:

$$c_1 = 0.3, \quad c_2 = 0.9, \quad c_3 = 0.9, \quad c_4 = 0.3, \quad c_5 = c_6 = -1.$$

Modeling Finite DC Gain Impacts

The tool substitutes ideal integration with a lossy approximation to predict leakage caused by finite amplifier gain ($\lambda = \omega_0/A_{vol}$):

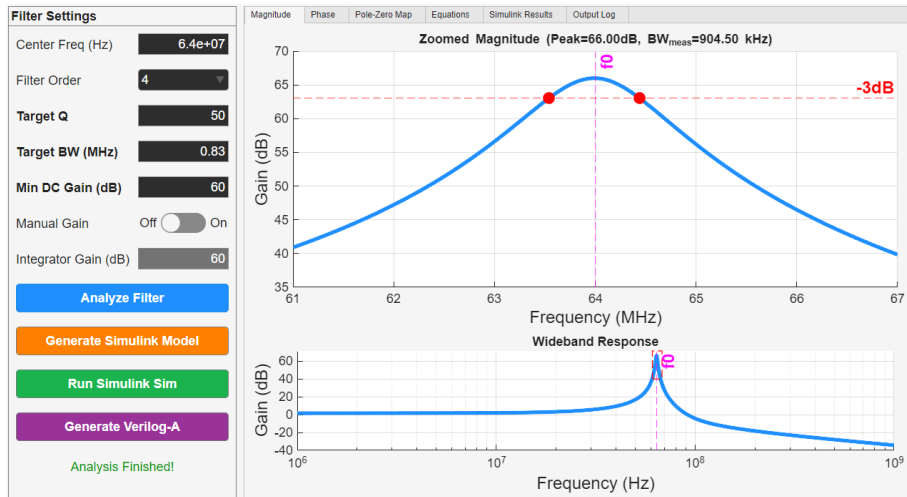
Lossy Integrator Transfer Function

$$H_{\text{real}}(s) = \frac{\omega_0}{s + \frac{\omega_0}{Q_{\text{target}}} + \frac{\omega_0}{A_{\text{vol}}}} \quad (1)$$

Resulting Q-Factor Degradation:

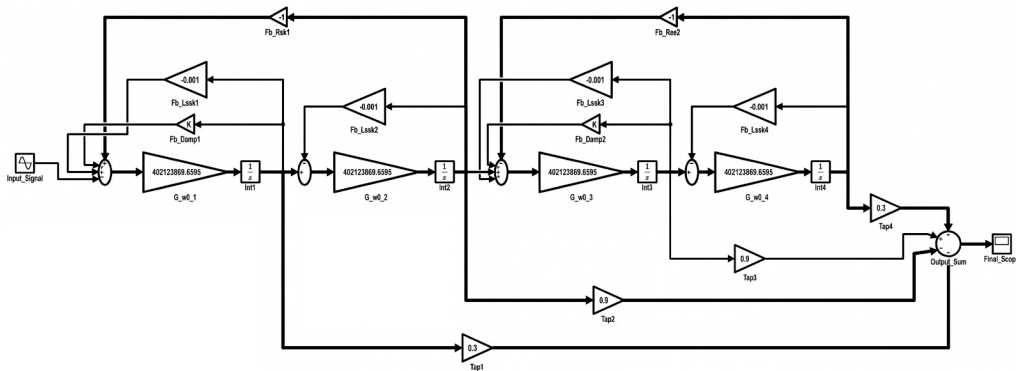
$$\frac{1}{Q_{\text{realized}}} = \frac{1}{Q_{\text{target}}} + \frac{1}{A_{\text{vol}}}$$

BETA Bandpass Design Tool | 4th-order CRFF Filter Synthesis



BETA Bandpass Design Tool | Simulink Framework for Verification

- A simulation environment is constructed within Simulink for verification.
- The tool handles the connection of all integration and gain coefficient blocks.
- Eliminates human errors, ensuring compatibility with the synthesis equations.
- The auto-generated **4th-order CRFF** filter architecture of is shown below.



BETA Bandpass Design Tool | Verifying the Simulink Model

- Transient and AC simulations are executed within Simulink to verify functionality.
- Time-domain waveform can be used to confirm stable, bounded signal propagation.
- The AC sweep demonstrates the bandpass characteristic centered at the target resonance frequency.
- Validates the synthesis equations before moving to transistor-level design.

Filter Settings

Center Freq (Hz)

Filter Order

Target Q

Target BW (MHz)

Min DC Gain (dB)

Manual Gain ☐ Off ☐ On

Integrator Gain (dB)

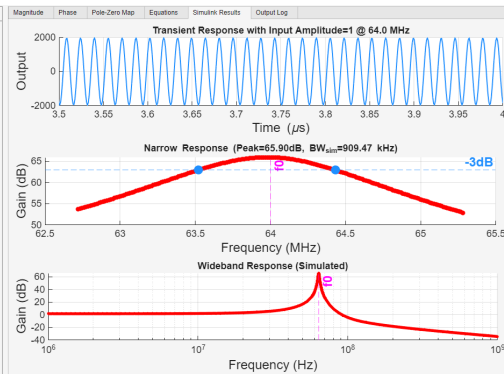
Analyze Filter

Generate Simulink Model

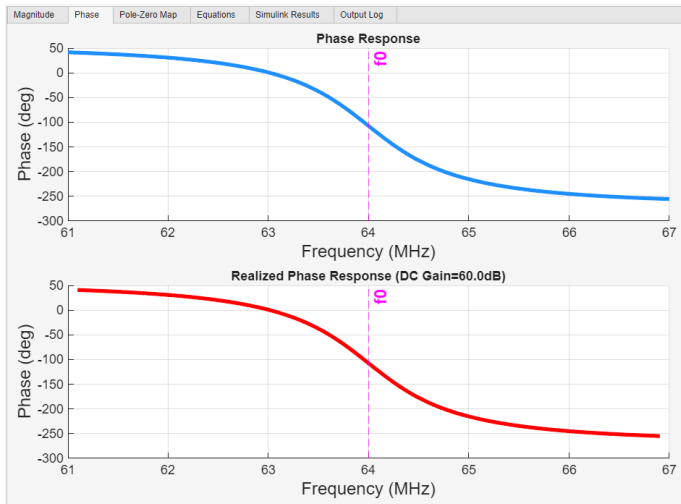
Run Simulink Sim

Generate Verilog-A

Simulation Done!



BETA Bandpass Design Tool | Analytical & Simulink Phase Response



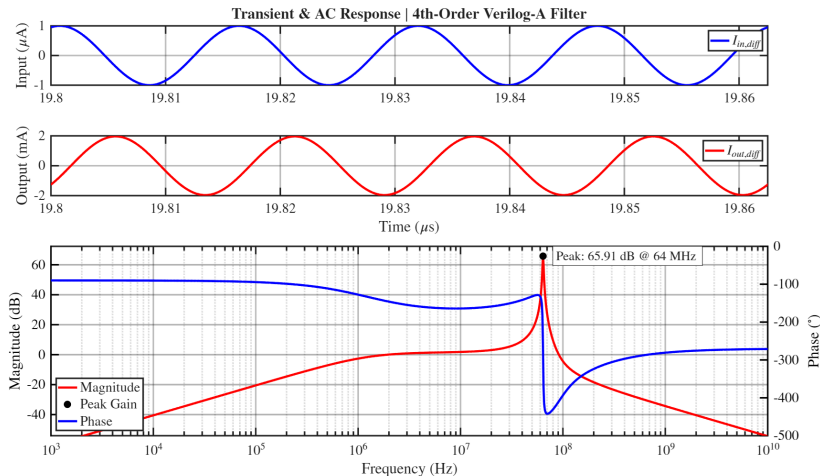
- A comparative phase response analysis is available to further evaluate modeling accuracy across the target band.
- Overlays the analytical behavior with the simulated outputs from the Simulink workspace.
- Verifies the accuracy of the automated transformation scripts used during synthesis.

BETA Bandpass Design Tool | Pole-Zero Mapping

- Pole-zero mapping highlights the stability margins of the 4th-order configuration.
- Tracks system poles & zeros under varying filter orders and DC Gains.
- Provides the guideline needed to place active stabilizing zeros within the loop.



BETA Bandpass Design Tool | Verilog-A Evaluation



- Verilog-A model of the integrator block can be generated for export into Cadence Virtuoso.
- Includes newly implemented Bandpass Verilog-A blocks for current-mode design.
- Establishes a reliable bridge between ideal mathematics and transistor-level design.

Transfer Function Testing | 1. Default Configuration

- **Custom TF: OFF**
- The tool relies on the precalculated default coefficients.
- Equal Q-factor distribution is automatically applied across the cascaded resonators.
- Provides a baseline, but lacks the flexibility needed for specialized bandwidth or dynamic range optimization.

System Parameters (4th-Order Default)

Center Freq (f_0): 10 MHz

Norm. Vectors: Num=[0.3, 0.9, 1.2, 1.2]
Den=[1, 0.04, 2, 0.04, 1]

Normalized TF:

$$H_{norm}(s) = \frac{0.3s^3 + 0.9s^2 + 1.2s + 1.2}{s^4 + 0.04s^3 + 2s^2 + 0.04s + 1}$$

Actual TF:

$$H_{act}(s) = \frac{1.88 \times 10^7 s^3 + 3.55 \times 10^{15} s^2 + 2.97 \times 10^{23} s + 1.87 \times 10^{31}}{s^4 + 2.51 \times 10^6 s^3 + 7.89 \times 10^{15} s^2 + 9.91 \times 10^{21} s + 1.56 \times 10^{31}}$$

Status: **Stable**

Expected System Q: 71.2

Expected BW: $f_0/Q_{sys} = 140.5$ kHz

Transfer Function Testing | 2. Unstable Configuration

System Parameters (4th-Order Custom)

Center Freq (f_0): 64 MHz

Norm. Vectors: Num=[0.3, 0.9, 1.2, 1.2]
Den=[1, -0.04, 2, -0.04, 1]

Normalized TF:

$$H_{norm}(s) = \frac{0.3s^3 + 0.9s^2 + 1.2s + 1.2}{s^4 - 0.04s^3 + 2s^2 - 0.04s + 1}$$

Actual TF:

$$H_{act}(s) = \frac{3.76 \times 10^7 s^3 + 1.42 \times 10^{16} s^2 + \dots}{s^4 - 5.02 \times 10^6 s^3 + 3.15 \times 10^{16} s^2 - \dots}$$

Status: **Unstable** (RHP Poles)

- **Custom TF: ON**
- User-defined transfer function vectors can be introduced to the tool.
- Demonstrates a scenario where improper coefficient selection (e.g., negative damping on the s^3 and s terms) pushes the system into instability.
- The poles migrate into the right-half of the s-plane, leading to an unbounded transient response.

Transfer Function Testing | 3. Stable with Adjusted Gains

System Parameters (4th-Order Custom)

Center Freq (f_0): 100 MHz

Norm. Vectors: Num=[0.3, 0.9, 1.2, 1.2]

Den=[1, 0.04, 2, 0.04, 1]

Int=[0.5, 2.0, 0.5, 2.0]

Normalized TF:

$$H_{norm}(s) = \frac{0.3s^3 + 0.9s^2 + 1.2s + 1.2}{s^4 + 0.04s^3 + 2s^2 + 0.04s + 1}$$

Actual TF:

$$H_{act}(s) = \frac{1.88 \times 10^7 s^3 + 3.55 \times 10^{15} s^2 + 2.97 \times 10^{23} s + 1.87 \times 10^{31}}{s^4 + 2.51 \times 10^6 s^3 + 7.89 \times 10^{15} s^2 + 9.91 \times 10^{21} s + 1.56 \times 10^{31}}$$

Status: **Stable**

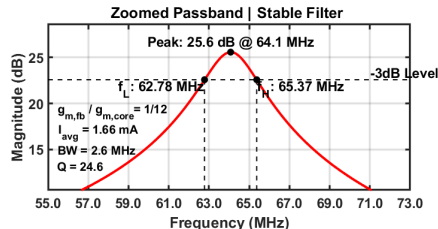
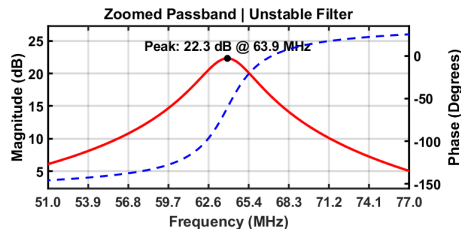
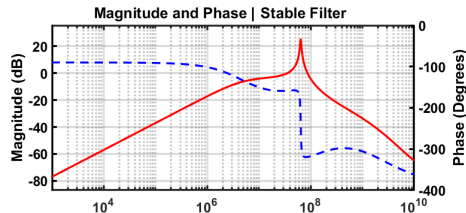
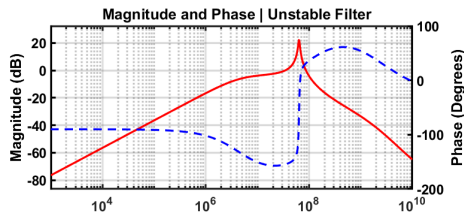
Expected System Q: 71.2

Expected BW: $f_0/Q_{sys} = 140.5$ kHz

- **Custom TF: ON (Stable)**
- **Different Integrator Gains** can be applied to optimize the internal dynamic range of the cascaded modulators.
- This could prevent internal node clipping while maintaining the target global transfer function and bandwidth.

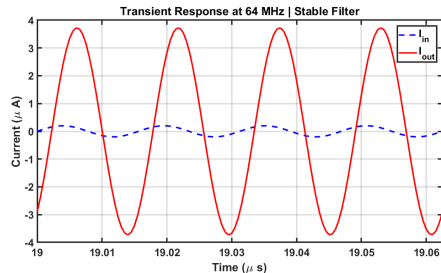
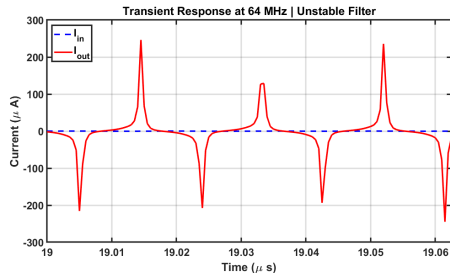
Transistor-Level Design | 2nd-Order CRFF Filter AC Response

- Confirms a well-behaved bandpass filtering characteristic centered at the 64 MHz target. (No Damping ie. $Q=\infty$ vs Target $Q=30$)



2nd-order Transistor-Level CRFF BP Filter: Unstable vs Stable

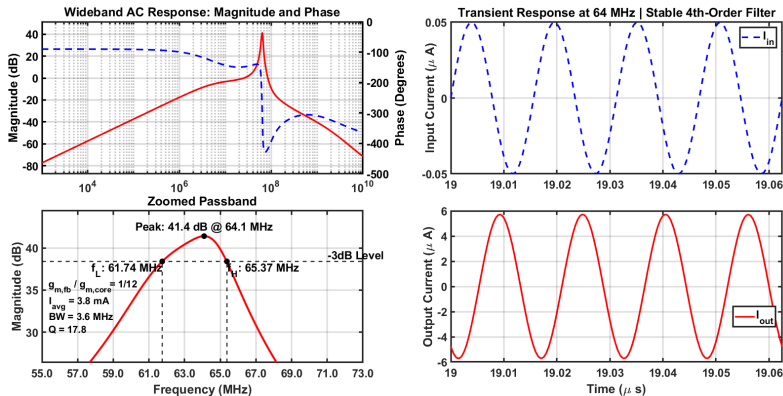
Transistor-Level Design | 2nd-Order CRFF Transient Response



- Compares the transient behavior of the unstable filter against the fully stabilized current-mode configuration.
- The unstable filter quickly exhibits rail-to-rail limit-cycle oscillations, rendering the filter non-functional.
- The stabilized configuration maintains bounded signal tracking across the time domain.

Transistor-Level Design | 4th-Order CRFF Filter Response

- The methodology is also validated on the **4th-order CRFF** loop filter. (Target $Q=20$)
- Verifies that the higher order block successfully maintain loop stability.



4th-Order Transistor-Level CRFF BP Filter Response

Conclusion

Impact on Design Cycle

The tool effectively links ideal synthesis with design environments, significantly reducing the iteration time for continuous-time high-order bandpass modulators.

High-order Precision

The custom normalized transfer functions enable a tangible way of analysis and design of a 4th-order bandpass filter (6th-order is an ongoing work).

Implementation Bridge

Automated Verilog-A and Simulink generation ensures that the designed architecture is verified before a transistor-level design.

Highlights of the MATLAB R2026b Prerelease

MATLAB

- **TOML Project File Format:** Manage projects in MATLAB and Simulink using a single, human-readable TOML configuration file.
- **MATLAB Package Management:** Share and install packages through File Exchange.
- **App Designer:** Save apps in plain text file format for use with source control.
- **Command Window:** Generate and display formatted text in the Command Window, using ANSI escape codes, Markdown text, and LaTeX equations.
- **Performance:** Key MATLAB data functions run faster for numeric and string arrays, tables, and timetables.

Simulink

- **Published Model:** Create ready-to-run versions of Simulink models you can reference to skip rebuilds and improve simulation performance.
- **Simulink Data Dictionary:** Save data dictionaries in plain text JSON files to facilitate source control diffs.
- **Simulation Data Inspector:** Configure tabs when visualizing simulation data to organize your layout and customize views.
- **Conditional Hold Block:** Use a single block to hold the value of an input signal based on a chosen conditional signal.
- **Simulink Profiler:** Identify simulation bottlenecks faster with explanations of results and solver profile summaries generated with Simulink Copilot.

Key References

Design Tools & Modeling

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Bandpass Filters & Resonators

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Thank You

Questions?

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